ROBERTO H JACOB DA SILVA

Redmond, Washington, USA

Email: robertohjdasilva@hotmail.com | LinkedIn: www.linkedin.com/in/robertohjdasilva | Website: rjsilva.org

PRINCIPAL SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS & OBSERVABILITY

Inventor (29 U.S. Patents) | Author | Creator of Engineering with a Heart

Principal Software Engineer with 20+ years of experience designing large-scale distributed systems, observability platforms, and diagnostic infrastructure for cloud and enterprise environments. Architect of automated telemetry and trace analysis systems used to debug complex distributed workloads at scale. Inventor of 29 U.S. patents across distributed systems, networking, and cloud infrastructure, and author of the memoir 'Impossible Crossing'. Combines deep systems programming expertise with architectural leadership focused on developer productivity, automated diagnostics, and operational transparency.

CORE EXPERTISE

- Observability & Diagnostics
- Distributed Systems & Cloud Platforms
- Virtualization & Networking
- Systems Programming

PROFESSIONAL EXPERIENCE

Microsoft, **Principal Software Engineer**, 2022 – Present (Redmond, WA)

- Lead architect and primary developer of an automated observability and diagnostics platform for a distributed VM orchestration system used in large-scale CI/CD infrastructure.
- Designed and implemented an end-to-end distributed tracing infrastructure using OpenTelemetry, capturing gRPC communication, asynchronous coroutine execution, and scheduler operations through centralized trace-context propagation.
- Built a custom distributed trace analysis platform, including a high-performance trace database, ingestion pipeline, and developer UI enabling timeline visualization, span inspection, and automated debugging workflows.
- Developed automated pipeline post-processing that analyzes distributed traces immediately after CI test runs, detecting failures, instrumentation gaps, and performance regressions while identifying critical execution paths.
- Introduced historical performance analysis across test runs, automatically comparing executions against historical baselines using statistical analysis (min, max, average, standard deviation) to identify anomalies.
- Iteratively redesigned the trace ingestion architecture, evolving from early prototypes based on Grafana Tempo and Azure Storage discovery to a direct ingestion model using dedicated processing machines.

- Delivered major performance improvements, reducing ingestion and indexing time for large trace datasets from ~40 minutes to <10 minutes and reducing historical trace comparison latency from ~2 minutes to <5 seconds; the platform is now used by ~35 engineers for daily troubleshooting and performance diagnostics.
- Improved VM infrastructure reliability by enhancing the Virtual Machine Manager deployment pipeline, optimizing workload distribution across CPU cores, strengthening isolation using cgroups and namespaces, and resolving stability issues in VM lifecycle management.

Lenovo, **Principal Software Engineer**, 2014 **–** 2022 (Remote, USA)

- Architected hybrid cloud infrastructure and telemetry systems for enterprise server platforms; recognized as Lenovo Master Inventor for sustained innovation and patent contributions.

IBM, **Principal Software Engineer**, 2013 – 2014 (San Jose, CA)

- Developed automated testing and diagnostics infrastructure for distributed enterprise systems and led initiatives to improve system reliability and engineering productivity.

Nokia, **Staff Software Engineer,** 2003 – 2013 (Calabasas, CA)

- Designed distributed networking software and high-availability telecom infrastructure supporting large-scale carrier systems.

PUBLICATIONS, AUTHORSHIP & PATENTS

- Author – *Impossible Crossing: An Inspirational Journey of Hope, Perseverance, and the Pursuit of Dreams (Amazon, 2022)*.
- Creator — Engineering with a Heart (https://rjsilva.org)

  Technical writing project explaining distributed systems, observability, and the hidden infrastructure that powers modern technology.

PATENTS & AWARDS

- Inventor of 29 U.S. patents spanning distributed systems, observability, cloud computing, and IoT.
- Lenovo Master Inventor; multiple engineering excellence awards.

EDUCATION

- B.S.E. in Electronics Engineering — Instituto Tecnológico de Aeronáutica (ITA), Brazil.